Stefanos Leonardos

Senior Lecturer in Machine Learning
King's College London, Department of Informatics

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Professional website ‖ **Personal website** ‖ **LinkedIn**



Personal: Born in Athens, Greece, 1981. Greek (native), German & English (fluent).

Expertise

- Artificial Intelligence & Machine Learning; Game Theory; Multi-agent Learning.
- AI for Financial Markets; Decentralised Finance; Cryptoeconomics.

Academic Experience

2022- **King's College London**
- Senior Lecturer in Machine Learning (Aug 2026 - present)
- Lecturer in Machine Learning (Jul 2022 - Jul 2026)

2018-22 **Singapore University of Technology & Design (SUTD)**
- Research Fellow: algorithmic game theory, dynamical systems, cryptoeconomics

2019-20 **University of Crete**
- Adjunct Researcher: innovation licensing, cross-ownership & taxation

2014-17 **National & Kapodistrian (NK) University of Athens**
- Doctoral Researcher: optimisation of stochastic systems under partial information

Education

PhD **Game Theory** — NK University of Athens, 2017
- Excellent. Supervisor: Costis Melolidakis

MSc **Statistics & Operations Research** — NK University of Athens, *2013
- Valedictorian 10/10

Diploma **Mathematics** — Technical University of Darmstadt, 2009
- Distinction 1.0/1.0

BSc, Hons **Business Information Systems** — Athens University of Economics & Business, 2004

*Compulsory military service: 2009-2010

Research Funding

2025-26 • **UKFIN+ network (UKRI)**, Principal Investigator, with **TurinTech**
£100,000: "SAiFE: a sandbox for AI-powered trading in decentralised financial markets".
Outputs: [**arXiv**], [**ICAIF**], Oct 2025 - Sep 2026.

2025-26 • **Exponential Science Foundation**, Science Fellow
$20,000: "Non-fungible token (NFT) based rewarding mechanisms".
Output: [**arXiv**], Oct 2025 - Sep 2026.

Leadership & Service

2026- Knowledge Representation, Reasoning and Multi-Agent Systems: research group head

2024- **King's Academic Board**: elected member

2022- **MAASAI Research Lab**: founder & lead
- King's affiliations: Economics & **Finance Hub** (co-lead, 2025-26); King's Institute for AI.

Area Chair: **ICML**, **ICLR**, **NeurIPS**, **AAAI**, **ECAI**.

Committees: **IJCAI**, **ACM EC**, **AAMAS**, **FC**, **AFT**.

Editor: [**ACM DLT**] Special issue on mathematical research for blockchain economy, 2022
[**Springer Nature**] **Marble** conference proceedings, 2023-2026 (Programme Co-Chair)

Examiner: L. Liu, Technical University of Munich (PhD); D. Belcher, University of Southampton (MPhil)

Organiser: **KCL Blockchain** Research Seminar, **Algorithmic Learning in Games** Seminar, **Informatics Research Showcase**, Informatics Departmental Seminar.

Awards & Recognition

2026 King's Education Award nomination
2025 Outstanding **Industry Engagement Award**: AWS Impact Accelerator module **(BRILLIANT 2025)**
2024 Fellow of **Advance HE** **(FHEA)**
King's Education Award nomination
2023 Featured Article (Editors' Choice): ACM Transactions on Economics and Computation **(ACM TEAC)**
King's Recognition Award
2021 Spotlight Paper: 35th Conference on Neural Information Processing Systems **(NeurIPS-21)**
Best Paper Award: IEEE Transactions on Engineering Management **(IEEE-TEM)**
Outstanding Paper Award: 35th AAAI Conference on Artificial Intelligence **(AAAI-21)**
2020 SUTD Outstanding Teaching Assistant Award
Best Paper Award: 3rd Annual Decentralized Conference
2019 Best Paper Award: 1st IEEE Conference on Blockchain and Cryptocurrency **(IEEE-ICBC)**
Best Poster Award: SUTD Research Fest
Best Paper Award: 1st Conference on Mathematical Research for Blockchain Economy **(MARBLE)**

2017 Doctoral scholarship (awarded; declined): **Greek State Scholarships Foundation** **(IKY)**
2013 Doctoral scholarship: **Alexander S. Onassis Public Benefit Foundation**

Teaching & Academic Mentoring

Finance (PGT):	Quantitative Methods; Scientific Computing; Agent-Based Modelling (co-lead, 50 students)
Machine Learning (PGT):	Pattern Recognition, Neural Networks & Deep Learning (co-lead, 200); AWS Impact Accelerator (80); Algorithmic Game Theory (SUTD, 20 students)
Personal Tutor:	40+ undergraduate & postgraduate students
MSc Supervision:	15 theses per year, Data Science MSc & Computational Finance MSc
PhD Supervision:	12 students: 6 as first & 6 as second supervisor; 3 currently at the defence stage

Publications (Selection)

Full list: **Google Scholar** ‖ **DBLP** ‖ **OpenReview**
• 1,518 citations, h-index 17, i10-index 26 (Google Scholar, Oct 2026)

ICAIF 2026 *SAiFE-gym: model-based environments for automated market making with concentrated liquidity*
G. Chionas, C. Kleitsikas, **S. Leonardos** (**SL**), L. Sánchez-Betancourt, C. Ventre
AAAI 2026 *STAR-MARL: LLM-based sub-task curricula design*
(LAMAS) Z. Li, D. Qiao, A. Rahman, **SL**, Y. Du, S. V. Albrecht
ECAI 2025 *From competition to centralization: the oligopoly in Ethereum block building auctions*
F. Wu, T. Thiery, **SL**, C. Ventre
arXiv 2025 *Performative market making*
C. Kleitsikas, **SL**, C. Ventre
ICLR 2024 *Beating price of anarchy and gradient descent without regret in potential games*
I. Sakos, **SL**, S. Stavroulakis, W. Overman, I. Panageas, G. Piliouras
AAMAS 2024 *Selfishness level induces cooperation in sequential social dilemmas*
S. Roesch, **SL**, Y. Du
NeurIPS 2023 *AlberDICE: addressing out-of-distribution joint actions in offline multi-agent RL*
D. E. Matsunaga, J. Lee, J. Yoon, **SL**, P. Abbeel, K.-E. Kim
ICLR 2022 *Global convergence of multi-agent policy gradient in Markov potential games*
SL, W. Overman, I. Panageas, G. Piliouras
NeurIPS 2021 *Exploration-exploitation in multi-agent competition: convergence with bounded rationality*
SL, G. Piliouras, K. Spendlove [Spotlight]
AAAI 2021 *Exploration-exploitation in multi-agent learning: catastrophe theory meets game theory*
SL, G. Piliouras [Outstanding Paper Award]

Invited Talks, Panels & Events (Selection)

2026 SIAM Conference on Optimization (OP26): Minisymposium on Optimization, Learning, & Games; King's College London: **AI for Finance Workshop** (co-organiser with C. Ventre; panellist)
2025 Isaac Newton Institute (**INI**): Reinforcement Learning; Algorithms, Learning & Games (**ALGA**); Continuous Optimization (**EUROPT**); House of Lords with Polkadot (host, Baroness Uddin); **Nash75**@Oxford
2024 Queen's University Belfast: Distributed Ledger Technology & QUB's Blockchain Initiative (panellist); Game-Theoretic Aspects in Blockchain Environments (**WINE**)
2023 University of Oxford; LSE; National University of Singapore (**IMS**); Algorithms, Learning & Games (**ALGA**)